

18/08/2026

$$\hat{H}\Psi = E\Psi$$

..... Eq. 1

Time-independent
Schrödinger
equation $\hat{H}$: Hamiltonian Operator Ψ : Wavefunction E : Total energy

Eq. 1 is an eigenvalue eq.

 $\hat{H}$ operator - hermitian & linear

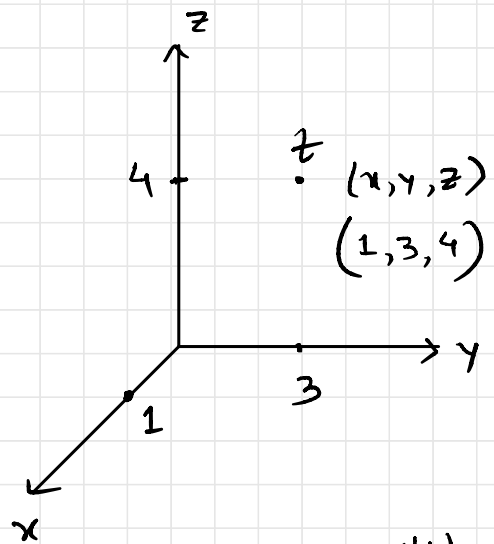
Observables Physical/classical

QM Operators (x-dimension)

	Symbols	Operations
1. x (position)	$\hat{x}$	Multiply by x -value
2. p_x (Momentum)	$\hat{p}_x$	$-i\hbar \frac{d}{dx}$ $i = \sqrt{-1}$ $\hbar = \frac{h}{2\pi}$
3. $V(x)$ (Potential energy)	$\hat{V}(x)$	Multiply by $V(x)$ -value
4. T_x (Kinetic energy) $= \frac{p_x^2}{2m}$	$\hat{T}_x$ $= \frac{\hat{p}_x^2}{2m}$	$-\frac{\hbar^2}{2m} \frac{d^2}{dx^2}$ $T_x = \frac{1}{2} m v^2$ $= \frac{p^2}{2m}$ $= \frac{\hat{p}_x^2}{2m} = \frac{\hat{p}_x \hat{p}_x}{2m}$ $= \left(-i\hbar \frac{d}{dx} \right) \left(-i\hbar \frac{d}{dx} \right)$ $= -\frac{\hbar^2}{2m} \frac{d^2}{dx^2}$ NOTE
5. E (Total energy)	$\hat{H} = (\hat{T} + \hat{V})$	$-\frac{\hbar^2}{2m} \frac{d^2}{dx^2} + V(x)$

Ex.

$$\hat{x} f(x) \\ \Rightarrow 1 \times f(x)$$

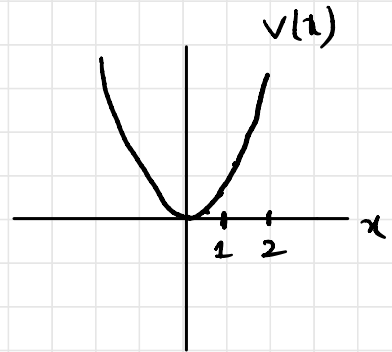


$$V(x) = \frac{1}{2} k x^2$$

$$\text{For } x = 2$$

$$\Rightarrow V(x) = \frac{1}{2} k 4^2 = 2k$$

$$\hat{V}(x) f(x) = 2k x f(x)$$



Properties of well-behaved/acceptable wavefunctions

1. finite
2. Continuous
3. single-valued
4. Continuous slope (differentiable)

For an acceptable wavefunction all four conditions to be satisfied!

